# Nicholas Richardson

## *Curriculum Vitae*

*Institute of Applied Mathematics*
*311-6356 Agricultural Road*
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*njericha.github.io*

## Research Interests

Audio/Signal Processing, Tensor Factorization, and Continuous Optimization

## Education

Sep. 2022 – (Jul. 2026) **Doctor of Philosophy**, *University of British Columbia*, Vancouver, BC
Institute of Applied Mathematics, supervised by Michael P. Friedlander
Thesis: *Computational Techniques for Analysis of Mixed Signals*

Sep. 2020 – May 2022 **Master of Mathematics**, *University of Waterloo*, Waterloo, ON
Applied Mathematics Department, supervised by Giang Tran
Thesis: *A Sparse Random Feature Model for Signal Decomposition*

Sep. 2016 – Apr. 2020 **Bachelor of Mathematics**, *University of Waterloo*, Waterloo, ON
Mathematical Physics Major, Pure Mathematics Minor

## Publications

2025 Joel Saylor, **Nicholas Richardson**, Naomi Graham, Robert Lee, and Michael P. Friedlander. Tracking cu-fertile sediment sources via multivariate petrochronological mixture modelling of detrital zircons. *Journal of Geophysical Research: Earth Surface*, 130, e2025JF008406.

2025 Naomi Graham, **Nicholas Richardson**, Michael P. Friedlander, and Joel Saylor. Tracing Sedimentary Origins in Multivariate Geochronology via Constrained Tensor Factorization. *Mathematical Geosciences*, 57, 601–628.

2023 Michael L. Waite and **Nicholas Richardson**. Potential Vorticity Generation in Breaking Gravity Waves. *Atmosphere*, 14(5):881.

2023 **Nicholas Richardson**, Hayden Schaeffer, and Giang Tran. SRMD: Sparse Random Mode Decomposition. *Communications on Applied Mathematics and Computation*, 6, 879–906.

2023 Lam Si Tung Ho, **Nicholas Richardson**, and Giang Tran. Adaptive group Lasso neural network models for functions of few variables and time-dependent data. *Sampling Theory, Signal Processing, and Data Analysis*, 21(2):28.

### Submitted

**Nicholas Richardson**, Noah Marusenko, and Michael P. Friedlander.
Multiple Scale Methods for Optimization of Discretized Continuous Functions. arxiv.org/abs/2512.13993.

## Teaching

Sep. 2023 – Apr. 2026 **Instructor**, *University of British Columbia*, Vancouver, BC
- Intro to Computing: Python (Fall 2025, Winter 2026)
- Calculus I (Fall 2023, Winter 2025)

Sep. 2022 – Dec. 2024 **Teaching Assistant**, *University of British Columbia*, Vancouver, BC
- Calculus I (Fall 2024)
- Calculus II (Winter 2023)
- Calculus III (Fall 2022)

Sep. 2022 – Apr. 2024 **Mathematics Learning Centre Tutor**, *University of British Columbia*, Vancouver, BC

June 2020 – Dec. 2024 **Mathematics & Physics Tutor**, *Self-Employed*
High School and Undergraduate Levels

Jan. 2018 – Dec. 2021 **Teaching Assistant**, *University of Waterloo*, Waterloo, ON
- Mathematics of Music (Fall 2020, 2021)
- Special Topics in Mathematical Connections: Math & Music (Fall 2021)
- Complex Analysis (Winter 2020)
- Calculus 1 & 2 (Fall 2019, Winter 2018, 2020, 2021)

## Awards

### McMaster University

Sep. 2026 Britton Postdoctoral Fellowship

### Society of Industrial and Applied Mathematics

Jun. 2026 OP26 Student Travel Award
Jul. 2024 AN24 Student Travel Award

### Natural Sciences and Engineering Research Council of Canada

May 2023 Canadian Graduate Scholarship – Doctoral (330 awards nationwide)
May 2021 Canadian Graduate Scholarship – Master's (840 awards nationwide)
May 2019 Undergraduate Student Research Award (3 149 awards nationwide)

### University of British Columbia

May 2026 Graduate Student Travel Award
May 2022 Faculty of Science PhD Tuition Award
Sep. 2022 Four Year Doctoral Fellowship
Sep. 2022 President's Academic Excellence Initiative PhD Award

### University of Waterloo

May 2021 President's Graduate Scholarship
Aug. 2020 K.D. Fryer Gold Medal
June 2020 Dean's Honours List With Distinction
May 2019 President's Research Award

### (Awards Continued)

Jan. 2019 Paul Berg Memorial Award

Jan. 2019 Arthur Beaumont Memorial Scholarship

Sep. 2016 President's Scholarship of Distinction

Sep. 2016 Alumni @ Microsoft Entrance Scholarship in Mathematics

## Presentations & Posters

Jun 2026 **SIAM Conference on Optimization**, *Edinburgh, UK*
Multiscale Methods for Optimization of Discretized Continuous Functions (Oral).

Jan 2026 **SFU Operations Research Seminar**, *Burnaby, BC*
Optimization and Applications of Unsupervised Signal Demixing (Oral).

Dec 2025 **Geology Industry Training with Teck and BHP**, *Vancouver, BC*
Tucker-1 Demixing Model: Under the Hood (Oral).

July 2025 **Joint SIAM/CAIMS Annual Meetings**, *Montréal, QC*
BlockTensorDecomposition.jl: An Optimization Framework for Constrained Tensor Factorization (Oral).

July 2025 **CAIDA/TrustML Workshop — ICML Visitors @ UBC**, *Vancouver, BC*
Unsupervised Signal Demixing with Tensor Factorization (Poster).

July 2025 **Vision and Learning Workshop @ ICML**, *Vancouver, BC*
Unsupervised Signal Demixing with Tensor Factorization (Poster).

Dec. 2024 **CMS Winter Meeting**, *Richmond, BC*
Density Separation with Tensor Factorization (Oral).

Oct. 2024 **INFORMS Annual Meeting**, *Seattle, WA*
Density Separation with Tensor Factorization (Oral).

Sep. 2024 **West Coast Optimization Meeting**, *Vancouver, BC*
Density Separation with Tensor Factorization (Oral).

July 2024 **SIAM Annual Meeting**, *Spokane, WA*
Density Separation with Tensor Factorization (Oral).

Oct. 2023 **Biennial Meeting of SIAM Pacific Northwest Section**, *Bellingham, WA*
Non-negative Tensor Decompositions for Unsupervised Learning (Oral).

June 2021 **Canadian Applied and Industrial Mathematics Society**, *Waterloo, ON*
Sparse Random Wavelet Representation & Decomposition (Poster).
Runner up in annual poster competition

## Software

### Published Code & Packages

2026 UBC Thesis Typst Template, *Typst*
github.com/njericha/ubc-thesis-typst-template

2024–2025 MatrixTensorFactor.jl & BlockTensorFactorization.jl, *Julia*
github.com/MPF-Optimization-Laboratory/BlockTensorFactorization.jl

### (Software Continued)

2023 SedimentSourceAnalysis.jl, *Julia*
github.com/njericha/SedimentSourceAnalysis.jl

2022 Sparse Random Mode Decomposition, *Python*
github.com/GiangTTran/SparseRandomModeDecomposition

2017 Drasil, *Haskell*
jacquescarette.github.io/Drasil

### Computer Skills

Julia, Python, MATLAB, Quarto, Typst, LaTeX, Adobe Software, Haskell

## Journal Papers Reviewed

2025 SIAM Journal on Scientific Computing (1 review).

2023–2026 Springer's Journal of Optimization Theory and Applications (3 reviews).

## Community & Leadership

Sep. 2025 – (Jul. 2026) **SIAM Student Chapter**, University of British Columbia
- Co-organizer for the first UBC Applied Mathematics Meeting (Oct. 2025)

Sep. 2023 – (Jul. 2026) **Math Graduate Committee**, University of British Columbia
American Mathematical Society Graduate Student Chapter
- Co-organizer and Seminar Host (2024–Present)
- Social Convenor (2023–Present)

## Student Supervision

May. 2026 – (Aug. 2026) **Michael Yang**, *Incoming Master's Student Summer Project*, University of British Columbia
Moment Matching for Density Estimation.

May. 2026 – (Aug. 2026) **Mehdi Naami**, *Undergrad Student Research Project*, University of British Columbia
Block Descent and Factorization with Automatic Differentiation.

Sep. 2024 – Jan. 2025 **Noah Marusenko**, *Undergrad Student Research Project*, University of British Columbia
Multiple Scale Methods for Tensor Factorizations.

## Additional Research

May 2019 – Aug. 2019 **Computational Fluids Research Assistant**, *University of Waterloo*, Waterloo, ON, supervised by Michael L. Waite

May 2017 – Aug. 2017 **Software Engineering Research Assistant**, *McMaster University*, Hamilton, ON, supervised by Spencer Smith & Daniel Szymczak